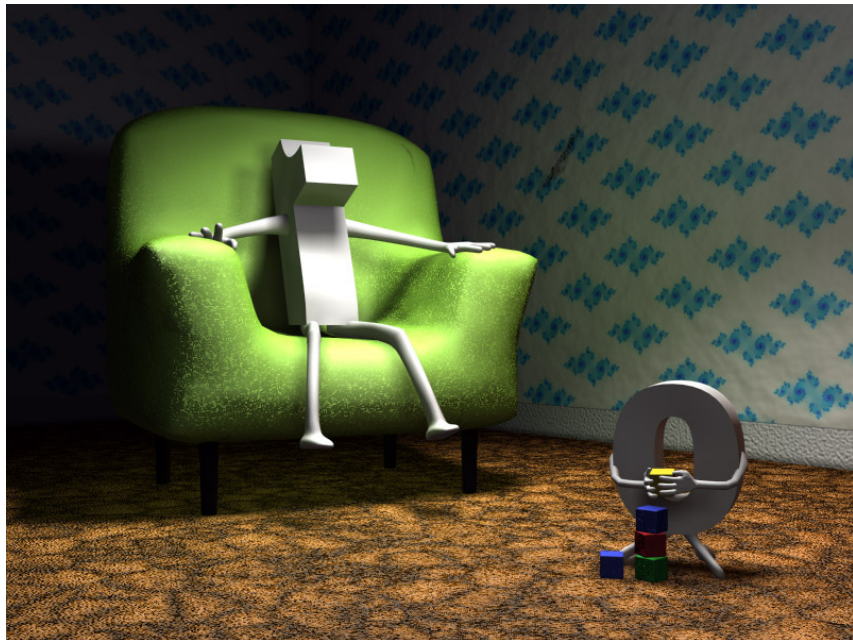


Chapter 1. INTEGERS

1. 1. Induction



In this short video we will see an example of how to apply the **PRINCIPLE OF MATHEMATICAL INDUCTION**.

PRINCIPLE OF MATHEMATICAL INDUCTION. Let p_n be a property or statement defined for each $n \in \mathbb{N}$. Suppose that the following two conditions hold.

- (1) **Basis Step:** p_1 is true.
- (2) **Inductive Step:** if p_k is true, then p_{k+1} is true. ($k \geq 1$)

Then the property p_n is true for every $n \in \mathbb{N}$.

Example 1.2. Prove by induction that

$$1 + 3 + 5 + \dots + (2n - 1) = n^2, \text{ for all } n \geq 1$$

Solution. Define p_n as

$$p_n : 1 + 3 + 5 + \dots + (2n - 1) = n^2.$$

Basis step. $n=1$ $p_1 : 1 = 1^2 \checkmark$

Inductive step: Suppose that p_k is true and let us show that p_{k+1} is also true.

$$P_{k+1}: 1 + 3 + 5 + \dots + (2k-1) + \underbrace{(2(k+1)-1)}_{2k+1} \stackrel{?}{=} (k+1)^2$$

$$\underbrace{1 + 3 + 5 + \dots + (2k-1)}_{\text{left hand side of } P_k} + (2k+1) \stackrel{P_k \text{ is true}}{=} k^2 + 2k + 1$$

$$= (k+1)^2,$$

which shows that P_{k+1} is true.

By the Principle of Mathematical Induction
we have that P_n is true for all $n \geq 1$

